

Programming Language Design: Constraints

Programming Languages Intro
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Constraints for Programming Language Design



Hardware Architecture



Technical settings



Standards



Legacy systems

Hardware Architecture



Most of the popular languages created are procedural or object-oriented

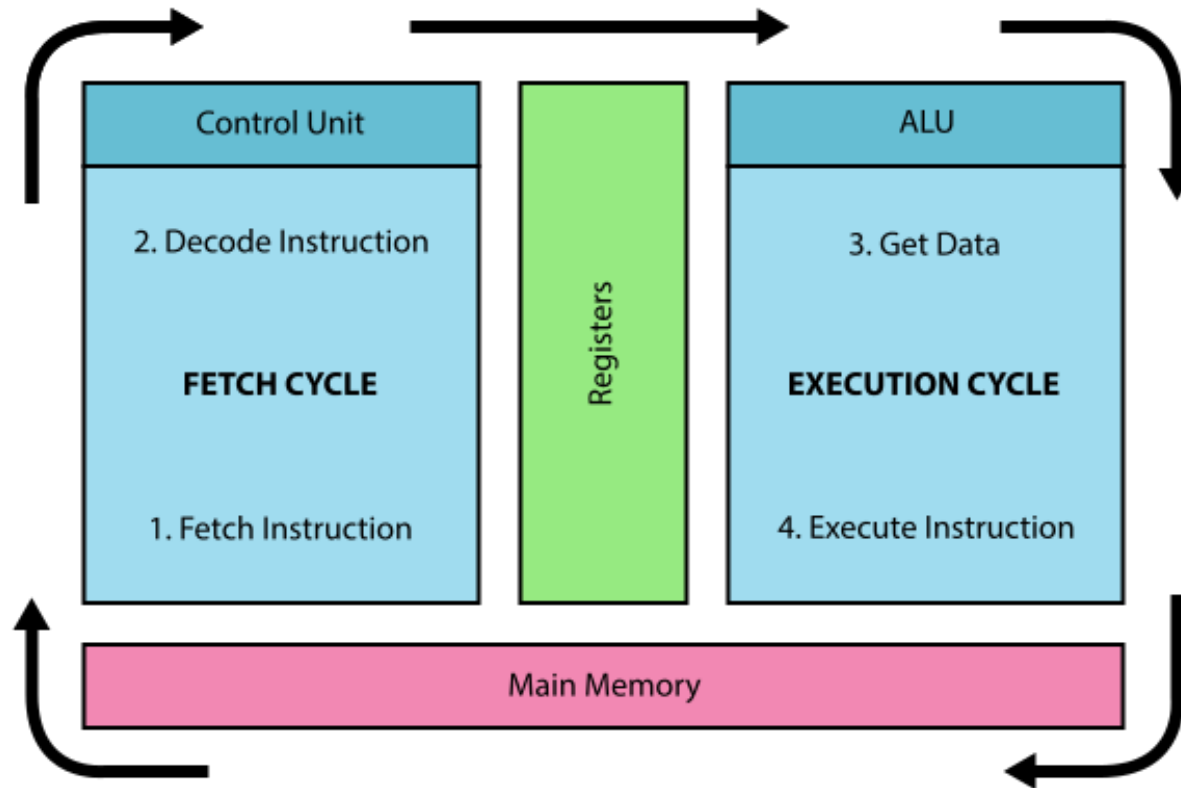


Why?

Von Neumann Architecture !

Hardware Architecture

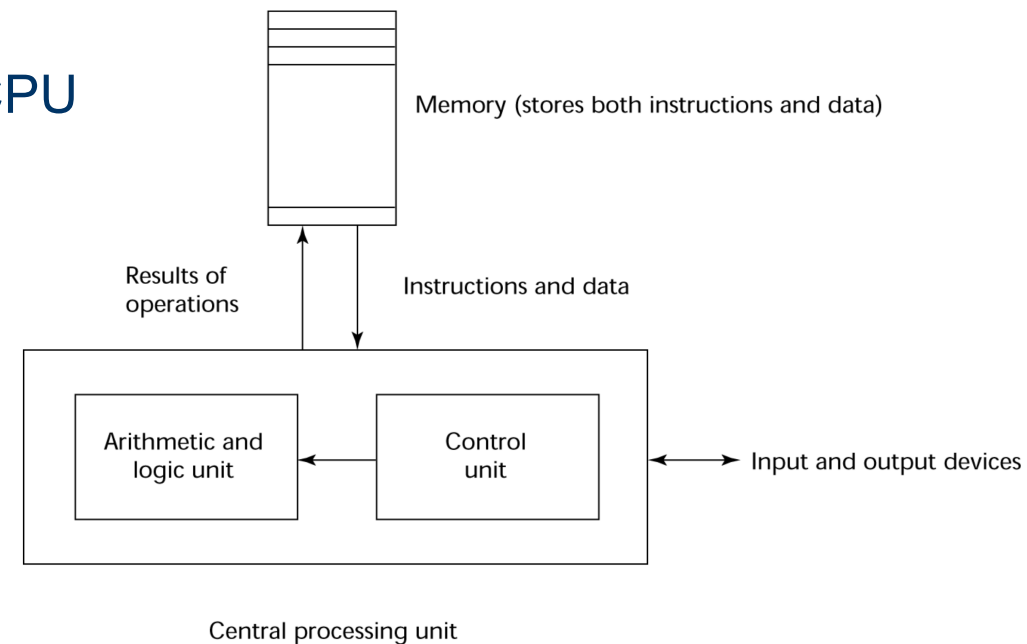
Von Neumann Architecture



```
initialize the program counter  
repeat forever  
    fetch the instruction  
    increment the counter  
    decode the instruction  
    execute the instruction  
end repeat
```

Hardware Architecture

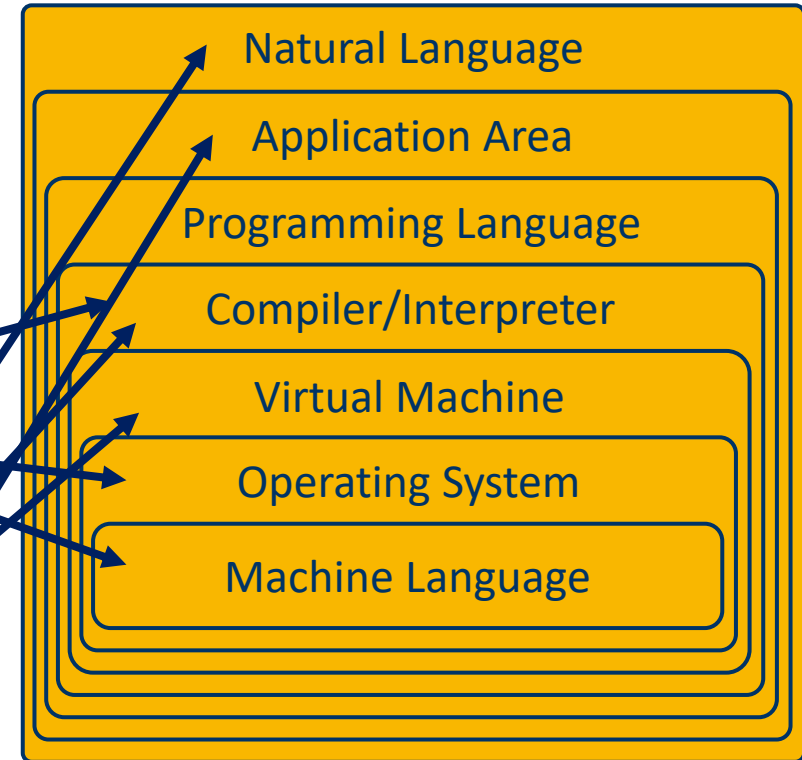
- Imperative languages are more efficient because the execution structure is similar
 - Data and programs stored in memory
 - Memory is separate from CPU
 - Instructions and data are piped from memory to CPU
- Basis for imperative languages
 - Variables model memory cells
 - Assignment statements model piping
 - Iteration is efficient
 - *Architecture discourages recursion*



Technical Settings

- Languages need to address the technical settings constraints where they will be used

FORTRAN is used in different platforms
with different compilers
adapting to the scientific programmers needs
These programmers use their own tools
and natural language to communicate



Technical Settings

- Languages have constraints like:
 - Application domain
 - Operating Systems and Hardware
 - IDE/Developer Tools
 - Preferences from the community or ecosystem

Language	Purpose
ADA	Address the needs of the US Department of Defense
COBOL	Support business-oriented applications
Prolog	Natural Language Processing, proving theorems, expert systems
C	Operating Systems development
PHP	Develop dynamic and interactive websites

Standards

- Focuses on availability on different platforms and groups
 - Portability
- Standards recognized organizations oversee and keep standards related to programming languages



Standards

- Some languages that went through standardization:
 - ISO/IEC 9899:2018 → C
 - ISO/IEC 14882:2020 → C++
 - ISO/IEC 1539-1:2018 → Fortran - Part 1: Base Language
 - ISO/IEC 1539-2:2000 → Fortran - Part 2: Varying length character strings
 - ISO/IEC 8652:2012 → Ada
 - ISO/IEC 30170:2012 → Ruby
 - ISO/IEC 1989:2014 → COBOL

ISO/IEC JTC 1

ISO/IEC JTC 1/SC 22

Programming languages, their environments and system software interfaces

Standards

Cons

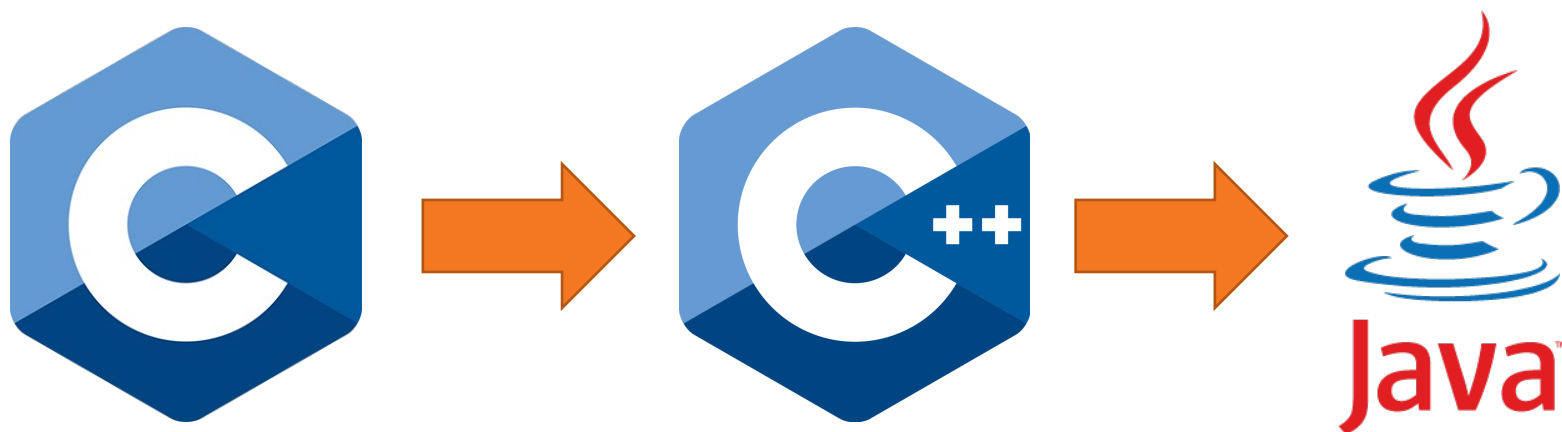
- Standardization is complex and slow
- Inhibits Innovation
- Poor characteristics are perpetuated

But...

- ISO and ANSI standards are revisited every 5 years
- Protect against obsolescence

Legacy System

- Support systems written in previous versions
 - Backwards compatibility
- Since previous resources cannot be removed
 - Languages become a bit overloaded



**See you in our next
Lesson!**